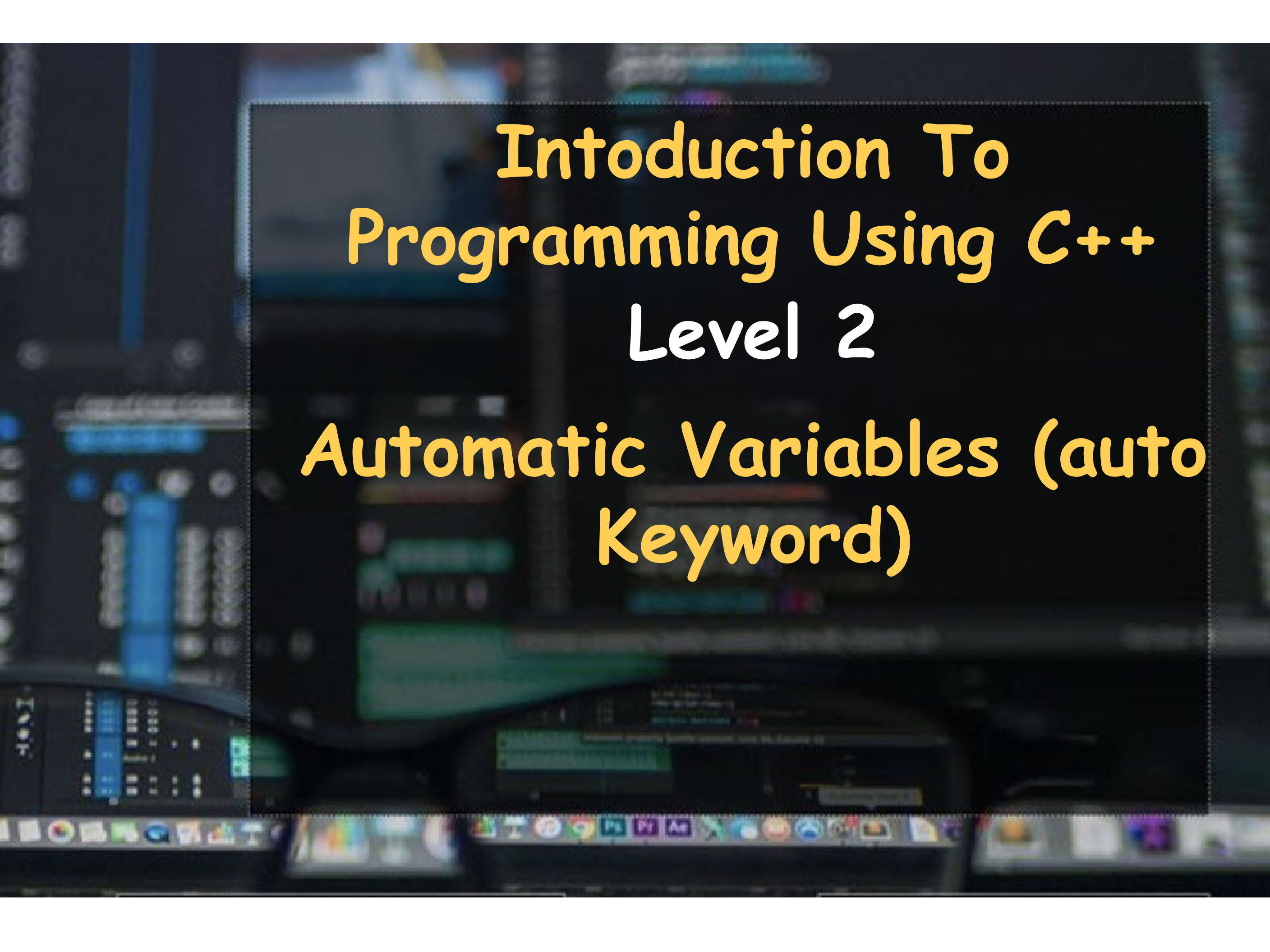
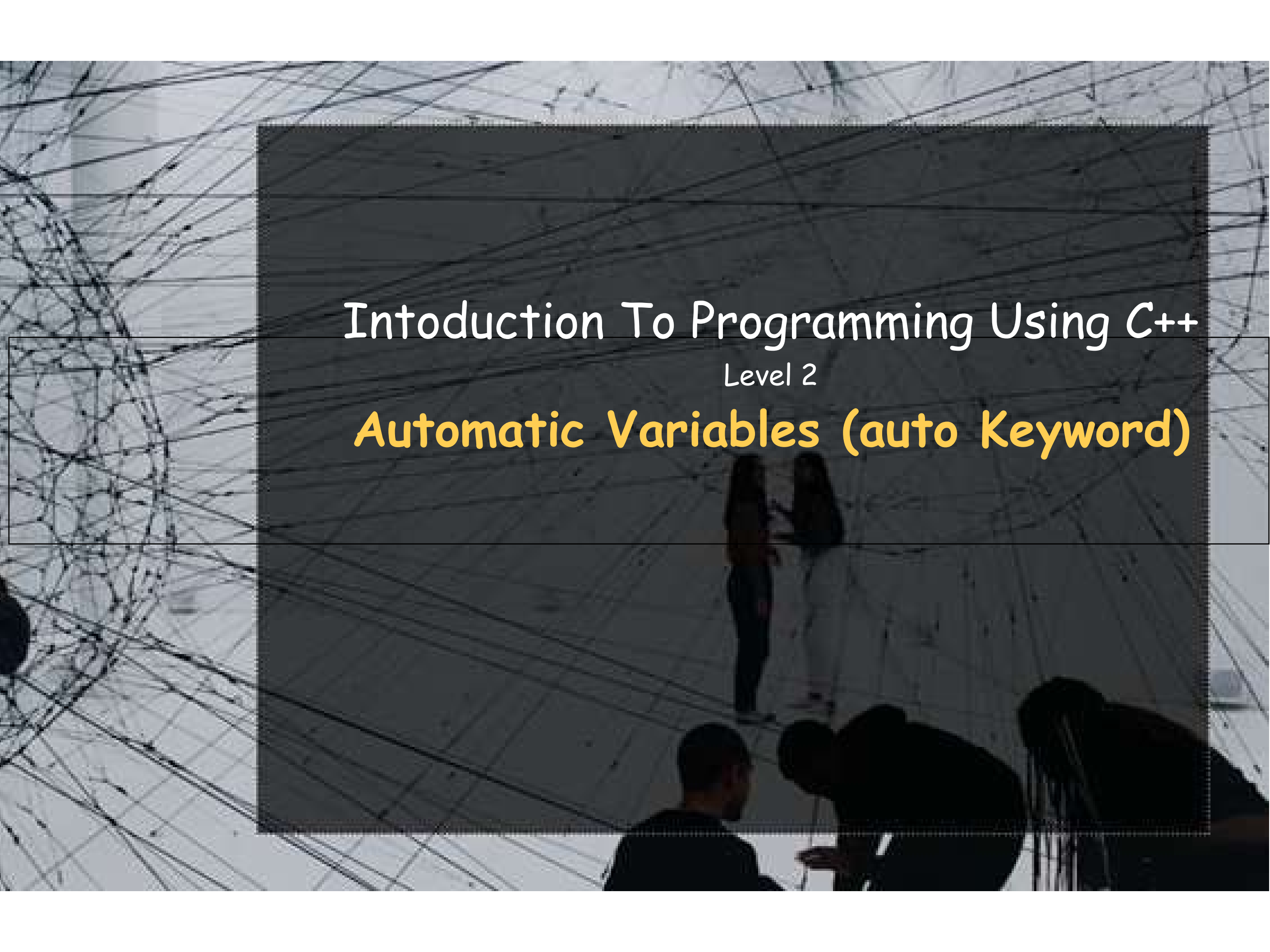




Introduction To Programming Using C++ Level 2

Automatic Variables (auto Keyword)

The background of the slide features a high-angle, black and white photograph of a modern building's interior. The ceiling is a complex, geometric structure made of many thin, intersecting lines, creating a web-like pattern. Below the ceiling, the silhouettes of several people are visible, some standing and some sitting, adding a sense of scale and activity to the scene.

Introduction To Programming Using C++

Level 2

Automatic Variables (auto Keyword)

Definition

- ◆ The ``auto`` Keyword allows the Compiler to automatically deduce the Type of a Variable from the value assigned to it at Compile Time.
- ◆ Instead of explicitly writing the Type such as ``int`` , ``double`` , or another Type, the Programmer can use ``auto`` , and the Compiler determines the appropriate Type automatically.

Explanation

- In C++, Variables normally require an explicitly specified Type when they are declared.
- With the `auto` Keyword, the Compiler determines the Variable's Type based on its initializer.
- This can make the code shorter and easier to read, especially when working with long or complex Types such as Iterators.
- The Type is determined at Compile Time, not while the program is running.

Procedures

- 1. Use the `auto` Keyword instead of explicitly writing the Type.
- 2. Assign a value to the Variable when declaring it.
- 3. The Compiler deduces the Type from the assigned value.

💡 Example

```
int main()
{
    auto x = 10;           // Type = int
    auto y = 12.5;         // Type = double
    auto z = "Mohammed Abu-Hadhoud"; // Type = const char*

    cout << x << endl;
    cout << y << endl;
    cout << z << endl;

    return 0;
}
```

➡ Result

10

12.5

Mohammed Abu-Hadhoud

Note

- ◆ `auto` does not mean that the Variable can change its Type later.
- ◆ Once the Compiler deduces the Type, the Variable has that Type normally.
- ◆ The Variable must be initialized when it is declared because the Compiler needs a value from which to deduce the Type.

⊘ Warning

- ◆ You cannot declare an `auto` Variable without an initializer:
 `auto a; // Error`
- ◆ The Compiler cannot determine the Type because no value was provided.



Comparison

▨ Using `auto`

- ✓ ``auto x = 10;``
- ✓ The Compiler deduces the Type automatically.
- ✓ Code can be shorter, especially with complex Types.
- ✓ The resulting Type and performance are determined at Compile Time.

▨ Using an Explicit Type

- ✓ ``int x = 10;``
- ✓ The Programmer specifies the Type directly.
- ✓ The Type is immediately visible to the reader.
- ✓ Can sometimes make the code clearer when the Type is important for understanding the logic.

✦ Summary

- ✓ ``auto`` allows the Compiler to automatically deduce a Variable's Type.
- ✓ The Type is determined from the initializer at Compile Time.
- ✓ An ``auto`` Variable must be initialized when it is declared.
- ✓ ``auto`` is especially useful for complex Types and can make code shorter.
- ✓ Using ``auto`` does not eliminate the need to understand Types.
- ✓ ``auto`` does not make a Variable dynamically change its Type..

Thank you 😊